Research on Machine Learning Methods for Predicting Risk of Cardiovascular Disease

1 Introduction

The purpose of this study was to explore the value of machine learning (ML) in cardiovascular disease risk prediction, and to determine the optimal prediction scheme by constructing and comparing multiple classification models (1). The core of this method is that the diagnosis of cardiovascular disease is essentially a typical dichotomous problem: judging whether a patient is at risk of cardiovascular disease based on their clinical characteristics. Traditional risk assessment methods rely on doctors' experience or simple scoring systems, while machine learning methods can automatically learn complex nonlinear relationships from multidimensional clinical data, to provide more accurate and objective predictions.

In terms of model selection strategy, this study adopted an methodological framework of multi-model comparison. The necessity of this choice can be seen in the following aspects: first, different machine learning algorithms have their own theoretical assumptions and applicability, for example, there is a linear relationship between the features of the logistic regression assumption and the logarithmic probability, secondly, by comparing various models, we can reveal the difference of the performance of different algorithms in processing medical data, finally, the multi-model comparison is also helpful to discover the intrinsic characteristics of the data set, such as the interaction between features and the degree of non-linear relationship. In this study, we hope to gain insight into the performance of the model, the importance of the features and the sensitivity of the hyperparameters, so as to provide scientific guidance for the actual deployment of the cardiovascular disease prediction system. Our GitHub link is https://github.com/songbo327/disease-analysis.

2 Problem statement

Cardiovascular disease is one of the leading causes of death in the world (2). According to the World Health Organization, about 17.9 million people die of cardiovascular diseases every year, accounting for 31% of all deaths worldwide (3). In this critical context, accurate early risk assessment of cardiovascular disease is critical for preventive medical interventions and the development of individualized treatment regimens. This study defined the cardiovascular disease prediction problem as a standard dichotomous task, that is, to judge whether a patient has cardiovascular disease based on multidimensional clinical features. Specifically, given patient records containing characteristics such as age, sex, height, weight, blood pressure, cholesterol levels, blood glucose levels, and lifestyle factors (such as smoking, alcohol consumption, exercise habits, etc.) , the model outputs a dichotomous label indicating whether the patient is at risk for cardiovascular disease.

The importance of this issue is not only reflected in its great public health value, but also closely related to clinical practice. From an academic research perspective, cardiovascular disease prediction spans many interdisciplinary fields such as epidemiology, biostatistics, and computational medicine, its findings can provide theoretical support for identifying disease risk factors and developing intervention strategies. From the perspective of industry application, accurate prediction models can be integrated into electronic health record system to assist doctors in clinical decision-making and realize the transition from passive treatment to active prevention. In terms of actual impact, early identification of high-risk groups can significantly reduce medical costs, to improve patient quality of

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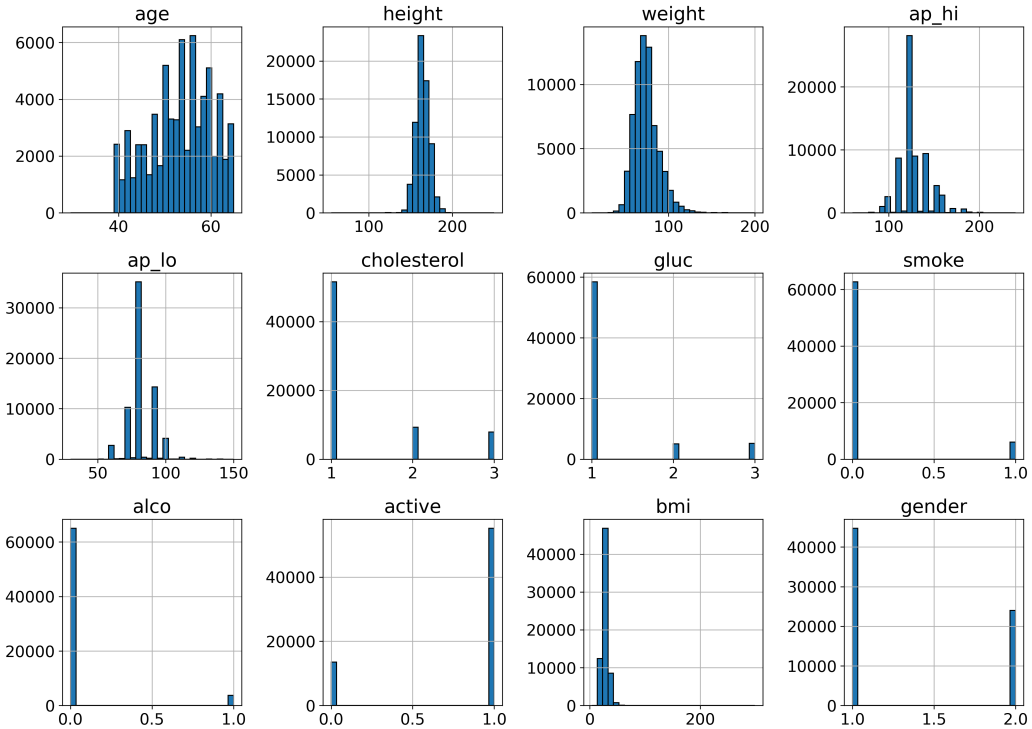

Figure 1: The histogram of feature distributions of the cardiovascular disease dataset.

life, and reduce the overall burden of the health system. Therefore, this study has not only theoretical values, but also social and practical significance.

3 Data Collection and Preprocessing

We use the data derived from the Kaggle platform-published cardiovascular disease dataset (sulianova/cardiovascular-disease-dataset), which was originally collate from patient checkup records collected by healthcare institutions and contains a wealth of clinical characteristics. The openness and widespread use of the dataset ensured reproducibility and comparability of the results. The original dataset contained a total of 70,000 samples, each with 13 attributes, including 1 identifier, 11 characteristic variables, and 1 target variable. The characteristic variables can be divided into three categories: objective fact-type features, including age, sex, height and weight, medical examination-type features, including systolic and diastolic blood pressure, cholesterol, blood glucose levels; And subjective information, including smoking status, alcohol consumption, and physical activity levels. The target variable, cardio, is a binary indicator of whether the patient has been diagnosed with cardiovascular disease as shown in Figure 3.

A descriptive statistical analysis of the dataset revealed several important characteristics, as shown in Figure 1. In terms of sample number, 68,737 valid samples were ultimately retained after pre-processing, and the data volume was sufficient to support reliable model training and evaluation. In terms of characteristic distribution, the age variable was presented in years after conversion, with an average of 53.29 years and a standard deviation of 6.76 years, indicating that the sample was mainly concentrated in the middle-to-old age group; the distribution of height and weight variables was relatively reasonable, with averages of 164.36 centimeters and 74.12 kilograms, respectively; the systolic average was 126.62 mmHg and the diastolic average was 81.35 mmHg in the blood pressure variable, which was in the normal high range. In terms of category balance, the distribution of target variables presented good balance characteristics, with a disease-free sample of 34,721 (50.5%), a disease-affected sample of 34,016 (49.5%), and a category ratio of 1.02; this balanced distribution avoids potential interference of category imbalance problems in model training.

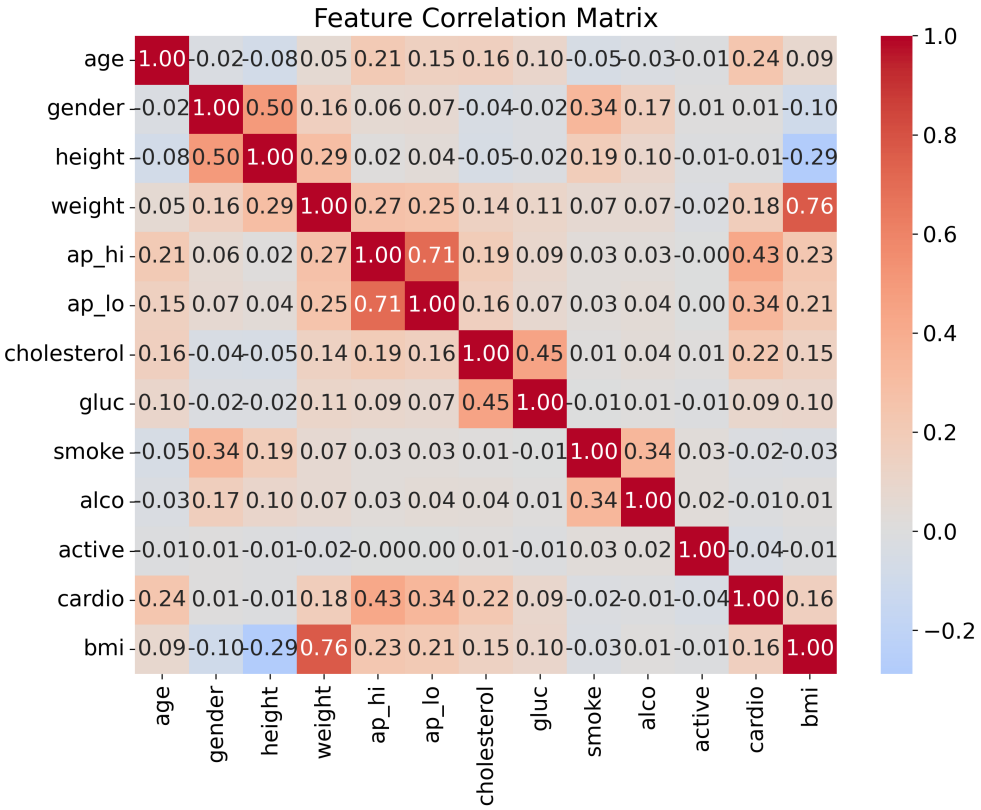

Figure 2: Correlation matrix of features.

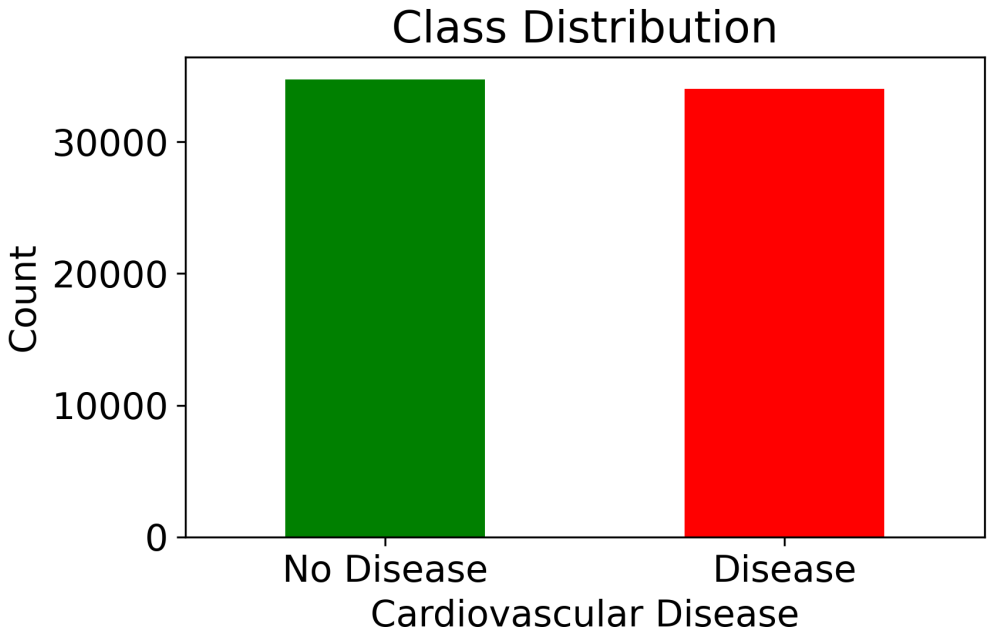

Figure 3: Class distribution of the cardiovascular disease dataset.

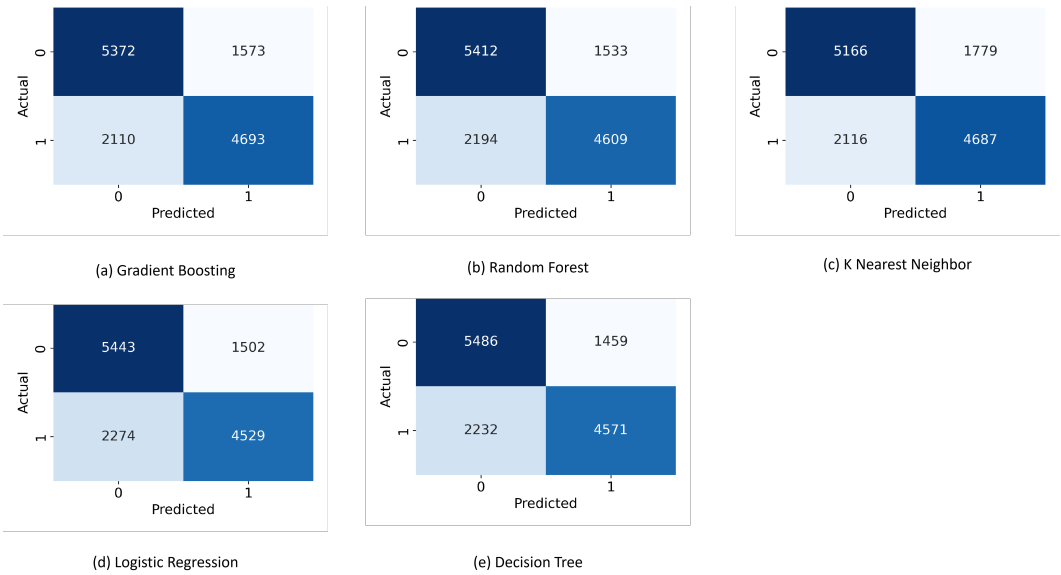

Figure 4: Confusion matrix of models.

In the data pre-processing stage, we adopt a systematic cleaning and conversion strategy to ensure data quality and validity of model training (4). First, we removed an identifier column that did not contribute to the prediction task. Secondly, the age variable is transformed from days to years, which improves the interpretability of the features. Third, the calculation of body mass index based on height and weight is an important derived feature used to assess the extent of obesity. The calculation formula of BMI is weight in kilograms divided by height in meters squared. Fourth, we performed duplicate record detection and deletion to ensure the uniqueness and integrity of the dataset. Fifth, outlier filtering was applied to blood pressure variables to maintain a reasonable range of systolic blood pressure between 50 and 250 mmHg and diastolic blood pressure between 30 and 150 mmHg, thus eliminating potential measurement or data entry errors. Together, these pre-processing steps ensure that subsequent modeling efforts are based on high-quality, high-reliability data.

4 Machine Learning Models Selection

In the process of model selection, the theoretical characteristics, computational complexity, interpretability and applicability of the model to medical data are comprehensively considered. After careful evaluation, we finally selected five representative machine learning models for comparative study, covering from linear to non-linear, from a single to integrated multiple model types (5).

As a classical linear classification model (6), logistic regression has been widely studied because of its solid theoretical basis, efficient computational performance and easy-to-interpret results. Logistic regression assumes that there is a linear relationship between the feature and the logarithm of probability, which is concise and performs well in many practical applications, especially in situations where the relationship is approximately linear. In addition, the coefficients of logistic regression can be directly interpreted as the extent to which features influence the predicted outcome, a feature that is particularly important in medical applications because physicians need to understand the decision-making basis of the model. As an example-based learning method, the K nearest neighbor classifier is chosen because of its non-parametric characteristics, that is, it does not make strong assumptions about data distribution and can adapt to various data patterns flexibly. The key idea of the algorithm is that similar samples should have similar labels by calculating the distance between the test samples and the training samples.

As a rule-based model, decision tree can automatically discover the interaction and non-linear relationship between features, the generated decision rules are widely studied for their high interpretability and easy for clinicians to understand and accept. As an integrated extension of decision tree, random forest (7) can reduce the risk of over-fitting and improve the generalization ability of the model by

constructing multiple decision trees and integrating their prediction results. Random forests also provide an estimate of the importance of features, helping to identify those features that contribute most to predictive performance. Gradient boosting tree (8) is another powerful integration method, which can reduce the prediction error by iterative training of weak learners, and has shown excellent performance in many machine learning competitions and practical applications. The advantage of gradient-boosting tree is that it can deal with feature interaction automatically, is more robust to outliers, and can balance the deviation and variance of model by adjusting learning rate and tree depth.

The main reason for this study to compare various models are that different models may show significant differences in processing medical data. Through systematic comparison, we can find the most suitable model for cardiovascular disease prediction task and reveal the intrinsic characteristics of the data set. The selection of evaluation indexes follows the specific requirements of medical application. In this study, four indexes are used: accuracy, F1 score, accuracy and recall rate. The overall prediction accuracy of the accuracy measurement model is the most intuitive performance metric; F1 score, as a reconciled average of accuracy and recall, can provide a more comprehensive performance assessment in the case of category imbalances; accuracy focuses on the proportion of samples predicted to be positive that are actually positive, corresponding to the control of false positives in medical diagnosis; and recall focuses on the proportion of samples predicted correctly that are actually positive, corresponding to the control of false negatives in medical screening, which is important for avoiding patients at high risk of misdiagnosis.

5 Model Development and Training

The model development phase employs a standardized machine learning process to ensure the scientificity of the experiments and the reproducibility of the results. First, the pre-processed dataset is divided into the training and test sets at an 80:20 ratio, with the training set containing 54,989 samples and the test set containing 13,748 samples. The division process employs hierarchical sampling strategies to ensure that the distribution of target variables in the training and test sets is consistent with the original dataset. Second, for models sensitive to feature scale, such as logical regression and K-nearest neighbors, standardized pre-processing is applied to convert the features to a normal distribution with an mean of 0 and a standard deviation of 1. For tree-based models, such as decision trees, random forests, and gradient enhancement trees, due to their inherent invariance, the raw features are used directly for training.

We used the grid search, a strategy that combines pentagonal cross-validation, to systematically explore the grid search space while evaluating the generalization performance of the model through cross-validation to avoid overfitting. For logistic regression, the normalized parameter C value of the search is [0.1, 1.0, 10.0], with a fixed maximum iteration count of 1000; for K-nearest neighbor classifiers, the number of neighbors of the search is [5, 10, 15], with a weighted strategy of ['uniform', 'distance']; for decision trees, the maximum depth of the search is [5, 10, 15], with a minimum split sample count of [10, 50, 100]; for random forests, the number of trees of the search is [50, 100], with a maximum depth of [10, 15, 20]; for gradient boost trees, the number of trees of the Search is [50, 100], the learning rate is [0.05, 0.1], with a maximum depth of [3, 5]. The grid search process evaluates the performance of each set of hyperparameter combinations under a framework of fifty-fifty cross-validation with accuracy as an optimized goal.

Table 1: Model comparison results

Model	Accuracy	F1_Score	Precision	Recall	AUC
Gradient_Boosting	0.7321	0.7182	0.7490	0.6898	0.7991
Decision_Tree	0.7315	0.7124	0.7580	0.6719	0.7875
Random_Forest	0.7289	0.7121	0.7504	0.6775	0.7987
Logistic_Regression	0.7253	0.7058	0.7510	0.6657	0.7893
KNN	0.7167	0.7065	0.7249	0.6890	0.7775

The experimental results (Table 1, Figure 3) showed that the gradient enhancement tree model achieved optimal performance on all evaluation metrics with an accuracy of 0.7321, F1 score of 0.7182, accuracy of 0.7490, recall rate of 0.6898, and AUC value of 0.7991. The decision tree

model followed closely with an accuracy of 0.7315, its accuracy being the highest among all models, reaching 0.7580, indicating that this model has high reliability in predicting patients with cardiovascular disease. The performance of the random forest model is slightly lower than that of the decision tree, with an accuracy of 0.7289, a result that may stem from the introduction of randomness in the random forest that reduces the model‘s determinism to some extent. Logical regression as a representative of linear models, with an accuracy of 0.7253, indicates that there is a certain linear relationship between features and target variables, but nonlinear models are able to capture more complex patterns. The accuracy of the K-nearest neighbor classifier is 0.7167, the lowest among all models, which may be because the distance measure in high-dimensional feature space is affected by the dimension curse.

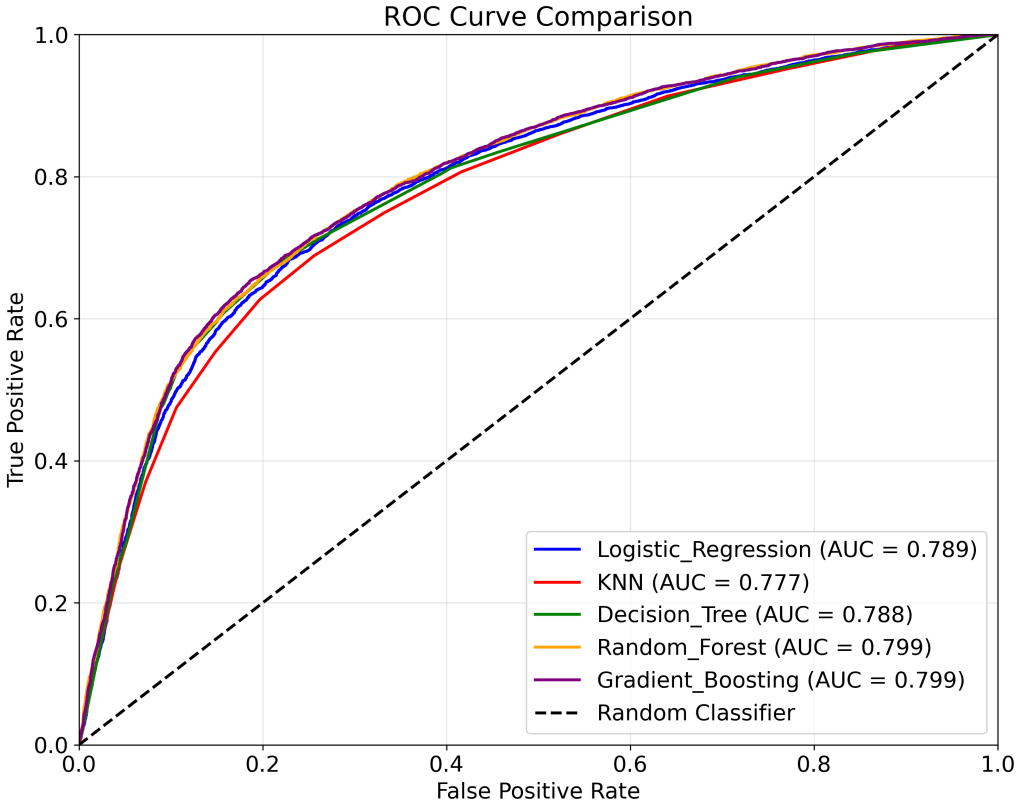

Figure 5: ROC curve comparison of models.

We performed a feature importance analysis based on a random forest model (Table 2), revealing the extent to which each feature contributes to cardiovascular disease prediction. Systolic blood pressure (ap_hi) ranked first with an importance score of 0.318, indicating that blood pressure level is the most critical predictor of cardiovascular disease, a finding that is highly consistent with medical common sense. Age ranked the second with an importance score of 0.174, reflecting the epidemiological regularity that the risk of cardiovascular disease increases with age. Diastolic blood pressure (ap_lo) ranked second with an importance of 0.138, which together with systolic blood pressure constitutes the overall contribution of blood pressure factors. The importance of the body mass index (BMI) was 0.104, highlighting the importance of obesity as a risk factor for cardiovascular disease. The importance of body weight and height was 0.077 and 0.067, respectively, the importance of cholesterol levels was 0.066, and the importance of blood sugar levels was 0.017. Notably, the importance of lifestyle factors such as physical activity, smoking, and alcohol consumption was relatively low, being 0.012, 0.009, and 0.007, respectively, which may be because these factors indirectly affect cardiovascular disease risk by influencing physiological indicators such as blood pressure and body weight.

Table 2: Feature importance ranking.

Feature	Importance
ap_hi	0.3176
age	0.1744
ap_lo	0.1378
bmi	0.1040
weight	0.0773
height	0.0669
cholesterol	0.0655
gluc	0.0174
active	0.0123
gender	0.0107
smoke	0.0087
alco	0.0075

Confusion matrix analysis further reveals the predictive properties of each model as shown in Figure 4. The gradient-boosting tree model achieved a better balance in the identification of true and true-negative cases, with a relatively balanced number of false positives and false negatives. Although the decision tree model has the highest accuracy rate, its recall rate is relatively low, indicating that it tends to make conservative prediction, reducing false positives but increasing false negatives. In medical detecting, this characteristic needs to be weighed because false negatives can mean missed high-risk patients with potentially serious consequences. The confounding matrix of logistic regression model showed a similar pattern, and the number of false negatives was slightly higher than that of false positives. The number of false positives and false negatives was closer in the K nearest neighbor model, but the overall accuracy was slightly lower.

6 Conclusions

Based on the above analysis, we constructed a set of machine learning model system for cardiovascular disease risk prediction, through systematic model comparison and in-depth feature analysis, providing a data-driven solution for early screening and risk assessment. The gradient lifting tree model is the first choice because of its excellent comprehensive performance, among which systolic pressure, age and diastolic pressure are the most important predictive characteristics. These findings not only validate known medical knowledge, but also provide quantitative evidence for risk factor interventions in clinical practice. Future studies could further explore in-depth learning methods, feature engineering techniques, and model interpretability enhancement to advance the application of cardiovascular disease prediction systems in real-world medical scenarios (9; 2; 10).

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